Azure Lighthouse Delegated Administration Design Study

Multi-Tenant Management

Course 24CC3046, Project Number P079 | My Team: T079 | 4 Students | Track: Microsoft Certified: Azure Administrator Associate

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Abstract

This design study tasks Team T079 (4 students) with architecting a cross-tenant management solution using Azure Lighthouse, modeling how a managed service provider (MSP) would administer multiple customer Azure Active Directory tenants from a single control plane. This is part of course 24CC3046, Project Number P079.

Core Objective

Design an Azure Lighthouse delegation model that lets a provider tenant manage resources in one or more customer tenants without requiring guest accounts or credential sharing in each tenant, while enforcing least-privilege access at every layer.

Key Design Areas Teams Must Address

- **Delegation structure** — ARM templates/Bicep for Microsoft Managed Services/registration Definitions and registration Assignments, mapping which provider Azure AD groups get which Azure RBAC roles in each customer tenant.
- **Least-privilege scoping** — Matching role assignments (e.g., built-in roles such as Contributor or Reader, or custom roles) to actual operational needs per resource group or subscription, avoiding over-broad Owner-level grants.
- **Cross-tenant visibility** — Using Azure Lighthouse's "My customers" view, Azure Resource Graph queries, and Azure Policy / Update Management across delegated tenants for consolidated auditing.
- **Audit trail consolidation** — Addressing the fragmentation problem where activity logs, sign-in logs, and policy compliance data live separately per tenant.

Stated Bottlenecks to Design Around

1. Cross-tenant permission scoping mistakes are high impact — a misconfigured role assignment can over-expose a customer's resources.
2. Auditing across tenants is fragmented — there is no native single-pane log aggregation across delegated tenants without extra tooling such as Azure Lighthouse combined with Log Analytics workspace federation or Microsoft Sentinel.

Deliverable Expectation

Typical for this certification-prep format: an architecture diagram, a delegation and role-mapping table, an ARM/Bicep onboarding template, and a written risk-mitigation section covering the two bottlenecks above.

Field	Detail
Course Code	24CC3046
Project Number	P079
My Team	T079
Team Size	4 students
Certification Track	Microsoft Certified: Azure Administrator Associate
Primary Focus	Cross-tenant management for a service provider model
Secondary Focus	Limiting delegated permissions appropriately